Identify Equivalent Algebraic Expressions

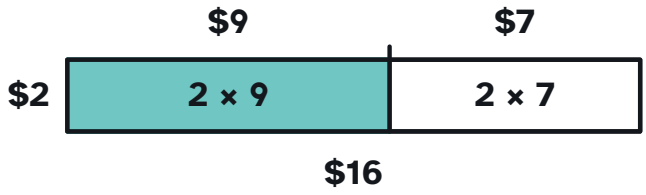
Test whether two expressions are equivalent by putting the same number into both.

Name _____ **Date** _____

Tonight's spark

A lunch combo is a sandwich that costs y dollars plus a $7 drink. Dani buys 2 combos and writes 2(y + 7). Kai writes 2y + 14. Do their costs match when y = 9?

Make a guess first: will the two costs match?



Our class model: test the same number in both

Try it

Work it out on paper. Write one answer on each line.

1 **What is the value of 3x + 1 when x = 4?**

A. 13 **B.** 16 **C.** 35

My answer _____

2 **What is the value of 2(x + 5) when x = 3?**

A. 11 **B.** 10 **C.** 16

My answer _____

3 **When x = 2, both 5x and x + 8 equal 10. What can you say?**

A. They are equivalent for every x **B.** They match at x = 2, so test another number

C. They are not equivalent because they look different **D.** They both equal 5

My answer _____

4 **Test x = 5. Do 2x + 4 and 6x give the same value?**

A. Yes, both are 30 **B.** Yes, both are 14 **C.** No, 2x + 4 is 30 **D.** No, they are 14 and 30

My answer _____

5 **Test n = 4 in 3(n + 2) and 3n + 2. What are the two values?**

A. 18 and 14 **B.** 14 and 14 **C.** 18 and 18 **D.** 12 and 14

My answer _____

6 **Which expression is equivalent to 2(m + 6)?**

A. 2m + 6 **B.** m + 12 **C.** 2m + 12 **D.** 8m

My answer _____

Talk about it

When the two values are different, what does that tell you? If they match once, are you finished?

Home mission

Pretend a sandwich costs y dollars and a drink costs $7. Pick y = 5. Ask your student to find 2(5 + 7) and 2 × 5 + 14. Do they match? Why can one number not be enough proof?

If we get stuck, ask

- What number will you put into both expressions?
- Did the parentheses change what happens first?
- One test disagreed — is that enough to decide?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/6-6-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it